

Three unit equilateral triangles in a square

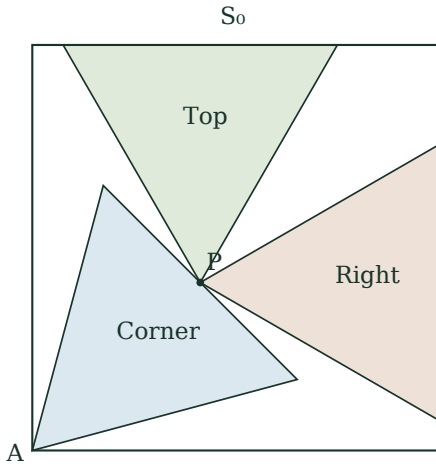
Computer-assisted proof candidate for review

Omer Zalman · Prepared for Erich Friedman · October 7, 2026

Proposed minimum: $S_0 = \sqrt{3}/2 + \sqrt{6}/4 \approx 1.47839784$

The triangles have side one and disjoint interiors; boundary contact is allowed. The exact construction and family bound below are combined with a recorded global computational exclusion claim. External mathematical review is pending.

1. Exact feasible construction



Put $h=\sqrt{3}/2$, $q=\sqrt{6}/4$, $a=\sqrt{2}/4$, and $S_0=h+q$. In the square $[0, S_0]^2$ use:

Corner: $(0,0)$, $(q+a, q-a)$, $(q-a, q+a)$.

Right: (q, q) , $(S_0, q-1/2)$, $(S_0, q+1/2)$.

Top: (q, q) , $(q+1/2, S_0)$, $(q-1/2, S_0)$.

The illustration reuses rounded GPU coordinates. The exact identities, rather than the drawing, establish feasibility.

All edge lengths are one because $h^2=3/4$, $q^2=3/8$ and $a^2=1/8$. Containment follows from $h, q > 1/2$ and $h, q > a$. The line $x+y=2q$ separates the corner triangle from the other two. Through $P=(q, q)$, the normal $(-1/2, h)$ gives right/top signed vertex gaps $(0, -h, 0)$ and $(0, 1/2, 1)$, so their interiors are disjoint.

2. Exact bound for the surviving boundary family

Suppose a triangle has vertex $A=(0,0)$ and the other two have unit bases on the right and top walls. Write $u=S-h$, with inward apices (u, z_1) , (z_2, u) , where $1/2 \leq z_i \leq S-1/2$. In the final interval $(\sqrt{3}+1)/2 < S < \sqrt{3}$, both apex directions lie inside every admissible corner wedge. Thus the corner triangle's opposite edge, with outward unit normal $n=(n_x, n_y)$, $n_x, n_y > 0$, requires:

$$A_1 = n_x u + n_y z_1 - h \geq 0, \quad A_2 = n_x z_2 + n_y u - h \geq 0.$$

For the right/top pair, wall and non-facing axes have negative vertex gaps; only the two facing sloped normals can separate them. Hence at least one holds:

$$C_L = (1+\sqrt{3})u - z_1 - \sqrt{3}z_2 \geq 0, \\ C_R = (1+\sqrt{3})u - \sqrt{3}z_1 - z_2 \geq 0.$$

Eliminate the two independently sliding apex coordinates using:

$$n_x A_1 + \sqrt{3} n_y A_2 + n_x n_y C_L = (n_x + \sqrt{3} n_y)[u(n_x + n_y) - h], \\ \sqrt{3} n_x A_1 + n_y A_2 + n_x n_y C_R = (\sqrt{3} n_x + n_y)[u(n_x + n_y) - h].$$

Positive coefficients imply $u(n_x + n_y) \geq h$. Since $n_x^2 + n_y^2 = 1$, Cauchy gives $n_x + n_y \leq \sqrt{2}$. Therefore $S \geq h + h/\sqrt{2} = S_0$. This family bound allows arbitrary corner orientation and independent apices; it assumes neither a shared apex nor rigidity.

3. Global coverage and exclusion certificate

Attained minimum and complete profiles. Compactness gives an attained minimum; area gives $16S^4 \geq 27$ and $S > 1$. Pair contacts are absent, vertex-vertex, proper vertex-edge, or positive edge-edge overlap: $1+9+18+9=37$ modes. Relabelling and square symmetries leave 334 wall-triple representatives and $334 \times 37^3 = 16,918,102$ cases. Full profiles record all active walls/contacts; absent ones have strict slack. Endpoint and zero-overlap limits use their actual point-contact profiles.

Reductions without restricting real angles. Necessary angular relations have integer bounds in 30-degree units on unwrapped branches. Flooring preserves integer individual/difference bounds and equalities, so the finite relation test is necessary for every real solution. Rotations remain continuous. Inward component translations and rigid span changes $A \cos \theta + B \sin \theta$, with one-sided derivatives and curvature for ties, eliminate a full profile only if it admits a smaller square. Coverage assumes no nonsingular rigidity.

Continuous domains and sound subdivision. Two rational charts $R(t)$, $-R(t)$, with $c=(1-t^2)/(1+t^2)$, $s=2t/(1+t^2)$, $t \in [-1,1]$, cover each rotation. Eight chart combinations cover the ten variables: six centroid coordinates, three t parameters and S . Overlapping side shells, outward dyadic bounds, covering child unions, and the exact precision lift from 2^{24} to 2^{30} preserve all candidates. Uncertain arithmetic or precision limits retain boxes. Separately written checkers verify necessary contractions, contradictions, domain transitions and queue coverage.

Certified descent and analytic dispositions. A proposed motion has $S'=-1$. For a rotating supporting normal n , other local vertex b , centre displacement d and quarter-turn J , its gap derivative is:

$$g' = n \cdot (v_{\text{other}} - v_{\text{owner}}) + \omega_{\text{owner}}(Jn) \cdot (d+b) - \omega_{\text{other}}(Jn) \cdot b.$$

Only strictly checked active-wall/contact derivatives certify descent; absent contacts have slack in the full profile. Dyadic velocity witnesses have denominator 1024. Flexible or singular configurations are retained unless a sound exclusion or motion resolves them. The 8,257 analytic profiles use Section 2; an earlier shared-apex rule has $u(n_x + n_y) = h$ and the same bound for $S > 1$. The recorded search covers through 1.4783977 and an overlapping final interval to an upper enclosure of S_0 .

4. Recorded verification and conditional conclusion

Verification item (reports dated October 6, 2026)	Recorded result
Continuous geometry inferences	245,521,202
Rejections / necessary contractions	135,634,128 / 109,887,074
Rigid / supplied-motion removals	430,561 / 10,471,492
Linked generations / exact field identities	33 / 85
Unverified recorded claims / unresolved queues	0 / 0

If the profile classification, reductions and certificate coverage are sound, any hypothetical minimum below S_0 enters an audited domain. Necessary contractions and covering subdivisions preserve it; valid contradiction or descent certificates cannot dispose of it. The remaining analytic families require $S \geq S_0$. Together with the construction, this would establish global optimality.

Verification scope. Separately written GPU verifiers were developed in the same project; this is not external peer review or formal proof-assistant verification. Review must scrutinize the classification, reductions, singular/boundary treatment and transcript correspondence. Source and recorded reports are supplied separately; full replay also needs the large archival records. Geometry and algebra ran on GPU; CPU handled orchestration, files and typography.

Primary problem: erich-friedman.github.io/packing/triinsqu/

Evidence: certificate.json, mathematical-review.txt and independent GPU verifiers.